

Origin of Small Understory Beech in New York Northern Hardwood Stands

Ralph D. Nyland

ABSTRACT

I sampled understory beech with heights that were 4 ft tall or less to determine their origin in 22 northern hardwood stands in New York. The proportion of seedling-origin beech ranged from 5 to 75% across sites. However, in 20 stands, the majority of beech (58–91%) with heights of 2–4 ft tall had a root sucker origin. Coupled with other information from published sources, findings suggest four possibilities about beech regeneration: (1) seed production and/or early seedling survival may have a clonal linkage, with some genotypes producing little viable seed, or with few seedlings surviving after germination; (2) a prevalence of root suckers among the taller understory beech suggests their higher potential for long-term survival and development and limited shifts in the genetic makeup of that species within a stand; (3) at some level of stocking, taller understory beech root suckers may interfere with survival and development of seedling regeneration; and (4) because the taller understory beech has a root sucker origin at most sites, root suckers may just develop more rapidly than seedlings.

Keywords: beech, understory beech, beech regeneration

Seedlings versus Root Suckers

Foresters and ecologists have long known that American beech (*Fagus grandifolia* Ehrh.) regeneration derives from two sources: seedlings and root suckers (see Nyland et al. 2006). Suckers develop from adventitious buds that form in callus on shallow roots after some kind of injury (Jones and Raynal 1988). Although disturbance from logging and similar activities may increase the potential for root sucker development (Houston 2001), many unmanaged stands also have abundant root suckers (Ward 1961, Held 1980). These also derive from callus on shallow roots, apparently after injuries from natural causes (e.g., root injuries caused by animals or movement of stones due to freezing and thawing of the soil). Some anecdotal information suggests that root suckers also become more prevalent as the beech bark disease progresses in a stand, and particularly after salvage operations (Houston 2005, Koch and Carey 2005).

The proportion of small beech may vary by origin across locations (Ward 1961, Held 1983), with both seedlings and root suckers occurring in stands across the species' geographic range and with many different site conditions (Held 1983). However, studies of understory beech have found a greater importance of root suckers at sites with the most severe environmental conditions (Ward 1961).

Factors such as parent tree age, unfavorable weather events, and infertility of parent trees likely influence seed production, and the importance of these may differ from one site to another, even resulting in no seed production in some cases (Stalter 1982, Houle 1994). Given an important influence of environmental conditions on flowering and seed maturation, poor seed crops across a region in any year may simply reflect the vagaries of weather and related factors. A continued paucity of seedlings at any one site through time might reflect either a local environment poorly suited to seedling survival or genetic factors related to seed production among parent trees.

How these factors play out at mesic sites considered suitable for beech remains unclear.

Conditions in New York

The northern range of American beech extends to the boreal forest in southeastern Canada, with New York well within the species' natural geographic distribution (Tubbs and Houston 1990). It occurs there in northern hardwoods at many mesic sites never cleared for agricultural use, ranging from an occasional tree in some cases, to an important component at other stands. Past studies in New York indicate that root suckers often become dominant beneath openings caused by a stand disturbance (Houston 1975), suggesting that seedlings have little importance in perpetuating the species. However, the high shade tolerance of beech and its potential to regenerate and survive in shade (Logan 1973, Tubbs and Houston 1990) would implicate seedlings as a likely source of regeneration as well, at least on mesic sites.

To evaluate this potential, I sampled stands having beech in overstory positions and checked the advance regeneration to identify its origin. I wanted to determine the proportion of small beech starting as seedlings, and whether seedlings have become an important component in the understory of northern hardwood stands containing larger beech trees.

Methods

Data for the analyses came from 16 widely dispersed mesic sites in the central Adirondacks (near Newcomb, New York) and another six dispersed mesic sites south of Syracuse, New York. All had northern hardwood stands of an uneven-aged character and evidence of some past logging, although not within three to five decades before the sampling. I made no attempt to determine the overall density of understory beech at any site or the stocking of larger beech. Rather,

Received January 2, 2008; accepted April 3, 2008.

Ralph Nyland (rnyland@syr.edu), SUNY College of Environmental Science and Forestry, Syracuse, NY 13210. Eddie Bevilacqua of SUNY-ESF offered statistical advice. Several forestry students under the supervision of Amy L. Bashant helped to gather data for the project. Funding came in part from the McIntire-Stennis program and from the Cooperative Forestry Research Unit of the University of Maine.

Copyright © 2008 by the Society of American Foresters.

Table 1. Proportion of beech stems ≤ 4 ft tall having a seedling origin among New York northern hardwood stands.

Location ^a	Seedlings as percent of all beech ^b		
	≤ 4 ft tall	< 2 ft tall	2–4 ft tall
1*	5	0	9
2*	10	0	20
3*	20	30	10
4*	20	30	10
5*	20	30	10
6	20	33	9
7*	25	20	30
8	25	27	24
9*	29	40	17
10	30	29	30
11*	33	63	5
12	33	31	33
13*	38	45	30
14*	43	65	20
15	46	60	33
16*	48	60	35
17*	48	65	24
18	48	75	29
19*	55	67	42
20*	58	80	20
21*	60	83	25
22*	75	90	60

^a Locations sorted by increasing percent of seedling origin beech ≤ 4 ft tall. An asterisk identifies Adirondack sites.

^b A proportionality test revealed no significant difference between Adirondack and other sites for seedlings of any size class.

I wanted to know the relative proportion of seedlings among randomly selected small beech trees at sites where that species occurred in abundance within the understory.

The sample included 20 randomly selected advance beech at six sites and 40 at the other locations, with some of the data gathered by student research assistants. At each site we stopped at 1-chain intervals along a transect dissecting the stand, selecting the small beech nearest to each stopping point, and randomly changing direction after evaluating each sample tree. We excluded current-year germinants and suckers and limited the sample to trees no more than 4 ft tall. That allowed clear determination of origin by pulling back the organic layer and upper mineral soil around each one or by excavating a tree if needed. We categorized each tree as either a seedling or root sucker and measured the height. Because of the unequal numbers of samples across sites, I converted the tree counts to percentages and grouped the advance beech into ones less than 2 ft tall and those 2–4 ft tall. To illuminate any pattern across sites, I sorted stands by ascending order for the proportion of seedlings within the sample and show the results in Table 1.

Results

Among small beech 4 ft tall or less, the proportion with a seedling origin ranged from 5 to 75% across the 22 sites (Table 1). For those less than 2 ft tall, the proportion was none to 90%. By contrast, seedlings accounted for no more than 35% of the 2- to 4-ft tall beech in 20 of the stands. The other two sites had 42 and 60% of the 2- to 4-ft beech with a seedling origin.

Discussion

Data gathered at 22 sites in New York reveal dramatic differences among stands in the relative proportion of small understory beech originating as seedlings and root suckers. However, even at most sites having relatively high proportions of seedling-origin small

beech, root suckers were more common among trees 2–4 ft tall. Such evidence suggests that the long-term presence of beech in most New York northern hardwood stands will largely result from root sucker development, and this will mostly perpetuate the genotypes already present.

Coupled with other information about beech regeneration (US Forest Service 2005, Nyland et al. 2006) these findings suggest four possibilities worth further consideration with respect to beech regeneration in New York northern hardwoods. First, findings revealed great differences in abundance of seedlings among stands in both regions of New York. This suggests that some sites may have beech genotypes with a low potential for seed production or a low potential for seedling survival after germination. That would minimize seedlings as a potential source of new beech trees at some sites, but not at others. Second, given that resistance to the scale insect serving as the vector for beech bark disease has a genetic basis (e.g., Houston 2005, Koch and Carey 2005, Ramirez et al. 2007), then new trees of root sucker origin would have the same degree of susceptibility as the overstory trees. As a consequence, effects of the disease within a stand would likely change little through time. Third, despite the high shade tolerance of beech, presence of taller root suckers may reduce light below the threshold for adequate photosynthesis to insure seedling survival and development. Consequently, more beech seedlings might survive and develop in stands having only a low density of tall understory beech. Fourth, root suckers may just develop more rapidly than seedlings, because of their connection to roots of larger trees and the potential for translocation of carbohydrates to enhance growth of the root suckers. If so, removal of tall understory beech saplings might prove necessary for promoting the development of beech seedling regeneration, particularly if managers viewed that technique as one means for increasing the abundance of seedling beech with a potential resistance or tolerance to beech bark disease.

These possibilities await testing. Nevertheless, whatever causes wide differences in the proportion of seedlings among small beech in New York northern hardwood stands, root suckers dominate the beech understory in most cases. As a consequence, root suckers rather than seedlings will more likely grow into the main canopy beneath gaps caused by overstory disturbances.

Literature Cited

- HELD, M.E. 1980. *An analysis of factors related to sprouting and seeding in the occurrence of Fagus grandifolia in the eastern deciduous forest of North America*. PhD dissertation, Ohio University, Athens, OH. 112 p.
- HELD, M.E. 1983. Pattern of beech regeneration in the east-central United States. *Bull. Torrey Bot. Club* 110:55–62.
- HOUSTON, D.R. 1975. Beech bark disease: The aftermath forests are structures for a new outbreak. *J. For.* 73:660–663.
- HOUSTON, D.R. 2001. *Effect of harvesting regime on beech root sprouts and seedlings in a north-central Maine forest long affected by beech bark disease*. US For. Serv. Res. Pap. NE-717. 20 p.
- HOUSTON, D.R. 2005. Beech bark disease: 1934 to 2004: What's new since Ehrlich? P. 2–7 in *Beech bark disease: Proc. of the Beech bark disease symp.*, Evans, C.A., J.A. Lucas, and M.J. Twery (eds.). US For. Serv. Gen. Tech. Rep. NE-331.
- HOULE, G. 1994. Spatiotemporal patterns in the components of regeneration of four sympatric tree species—*Acer rubrum*, *Acer saccharum*, *Betula alleghaniensis*, and *Fagus grandifolia*. *J. Ecol.* 82(1):39–53.
- JONES, R.H., AND D.J. RAYNAL. 1988. Root sprouting in American beech (*Fagus grandifolia*): Effects of root injury, root exposure, and season. *For. Ecol. Manag.* 25(2):79–90.
- KOCH, J.L., AND D.W. CAREY. 2005. The genetics of resistance of American beech to beech bark disease: Knowledge through 2004. P. 98–105 in *Beech bark disease: Proc. of the Beech bark disease symp.*, Evans, C.A., J.A. Lucas, and M.J. Twery (eds.). US For. Serv. Gen. Tech. Rep. NE-331.

- LOGAN, K.T. 1973. *Growth of tree seedlings as affected by light intensity: V. White ash, beech, eastern hemlock, and general conclusions*. Department of Environment, Can. For. Serv., Publ. 1323. 11 p.
- NYLAND, R.D., A.L. BASHANT, K.K. BOHN, AND J.M. VEROSTEK. 2006. Interference to hardwood regeneration in northeastern North America: Ecological characteristics of American beech, striped maple, and hobblebush. *North. J. Appl. For.* 23(1):53–61.
- RAMIREZ, M., J. LOO, AND M.J. KRASOWSKI. 2007. Evaluation of resistance to the beech scale insect (*Cryptococcus fagisuga*) and propagation of American beech (*Fagus grandifolia*) by grafting. *Silvae Genet.* 65(3–4):163–169.
- STALTER, R. 1982. Production of viable seed by American beech (*Fagus grandifolia*). *Bull. Torrey Bot. Club* 109(4):542–544.
- TUBBS, C.H., AND D.R. HOUSTON. 1990. *Fagus grandifolia* Ehrh. American beech. P. 325–332 in *Silvics of North America, Vol. 2. Hardwoods*. US For. Serv. Agric. Handb. No. 654.
- US FOREST SERVICE. 2005. *Beech bark disease: Proc. of the Beech bark disease symp.*, Evans, C.A., J.A. Lucas, and M.J. Twery (eds.). US For. Serv. Gen. Tech. Rep. NE-331. 149 p.
- WARD, R.T. 1961. Some aspects of the regeneration habits of the American beech. *Ecology* 42(4):828–832.